`

AMRITA VISHWA VIDYAPEETHAM

AHEAD Online

Digital System Design

Veena S
Department of CSE
Amrita Viswa Vidyapeetham

Objective

- Brief Introduction

Logic circuit

Digital devices

Digital System Design

Transistors

Electrons

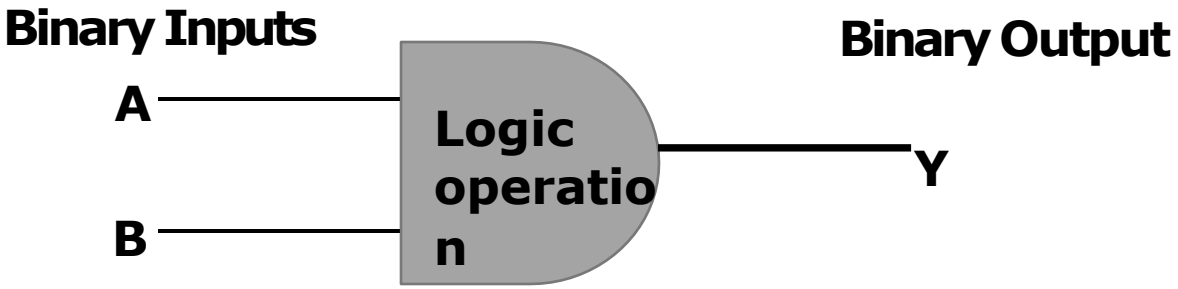
AMRITA VISHWA VIDYAPEETHAM | AHEAD Online

Why do we study Digital System Design?

- Digital design deals with the design of digital electronic circuits. e.g., microprocessor, RAM, ALU …
- Digital circuits operate using digital signals (usually use discrete value) in contrast to analog signals (continuous range of values).
- Digital systems use **binary logic** to store, process and manipulate data.

Binary Logic

- **Binary logic or Boolean logic** deals with binary variables (input and output) and logical operations.
- Binary variables are represented using alphabets A, B, x, y…
- Logical operations are functions applied to one or more binary input and produce a single binary output



1	0
TRUE	FALSE
HIGH	LOW
5V	0V

Logical operators

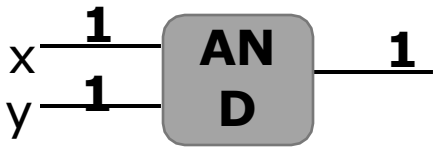
Binary variables : x , y, z with values 1 or 0

AND

Represented by dot (.)

$x \cdot y = z$ read as "x AND y equal to z"

$z = 1$ iff $x = 1$ and $y = 1$;
otherwise, $z = 0$



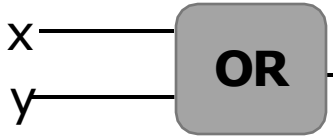
$z = x \cdot y$

OR

Represented by plus(+)

$x + y = z$ read as " x OR y equal to z"

$z = 1$ if $x = 1$ or $y = 1$ otherwise, $z = 0$



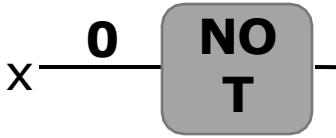
$z = x + y$

NOT

Represented by prime (')

$x' = z$ read as " not x is equal to z"

$z = 1$ if $x = 0$



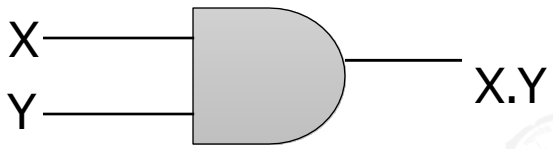
$z = x'$

Logic Gates

- We can realize each of the Boolean function using **logic gates**.
- Logic gates are the building blocks of all circuits in a computer.
  - Produces an output based on one or more inputs.
  - Some basic logic gates are AND, OR, NOT, NAND, NOR etc.
- We can represent the logical operations using truth table, Boolean expression, venn diagram, switching circuit.

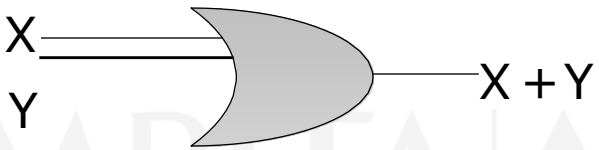
Logic gate

AND



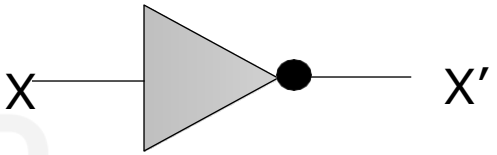
INPUTS		OUTPUT
X	**Y**	**X.Y**
0	0	0
0	1	0
1	0	0
1	**1**	**1**

OR



INPUTS		OUTPUT
X	**Y**	**X+Y**
0	0	0
0	**1**	**1**
1	**0**	**1**
1	**1**	**1**

NOT



INPUTS	OUTPUT
X	**X′**
0	**1**
1	0

AMRITA VISHWA VIDYAPEETHAM | AHEAD Online

Design a Digital System

- Understand the real-world problem.

  e.g., If out of 2 door switch (M1 and M2) any one is pressed a doorbell should be ringed.

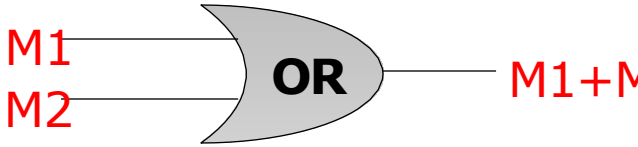
- Understand the truth table or logic behind the problem

M1	M2	Op
0	0	0
0	1	1
1	0	
1	1	

- Represent the truth table using a Boolean expression

  $Op = M1 + M2$

- Design the digital circuit i.e., hardware implementation



Summary

- We discussed about the basic concepts of digital logic.

Objective

- Boolean Algebra

Introduction

- Digital inputs and outputs are binary values i.e., 0 and 1.
- Binary operations performed by digital circuits are called logical functions or operations.
- Logic gates are the implementations of the logical operations.
- Multiple logic gates can be connected from output to input to design complex circuits.
- **Boolean algebra** can be used to reduce or simplify the complex circuits.

Boolean Algebra

- Mathematics of logic design, introduced by **George Boole** in 1854.
- **Boolean algebra** deals with logical operations on binary variables with a set of laws or rules.
- **Laws of Boolean algebra** can be used to reduce the number of logic gates by reducing the Boolean expressions.
- One application of Boolean algebra is in the analysis and design of digital systems.

Boolean Function

- Expression formed with binary variables, logical operators , parentheses and an equal sign.

  e.g., $\mathbf{F}(x, y, z) = \mathbf{x} + \mathbf{y.z'}$

  1 + 1

- Function can be either 1 or 0 based on the binary variable value.
- In the above e.g., F = 1 if x = 1 *and* y = 1*and* z′ = 1 otherwise F = 0.
- Function can be represented as a **Boolean expression** (same as e.g.,)or in a ***truth table*** *or using* ***logic diagram****.*
- Operator Precedence order: Parentheses, NOT, AND , OR

Truth Table

- Tabular representation of a Boolean Function.
- A truth table shows all the possible inputs and outputs of a function.
- A function with n variables has 2^n possible combinations of inputs.
- **Truth table of AND, OR, NOT with 2 inputs**

2 inputs - $2^2 = 4$ outputs

inputs		output		
x	**y**	**x.y**	**x +y**	**x'**
0	0	0	0	1
0	1	0	1	
1	0	0	1	0
1	1	1	1	

$F(x, y) = 2^2 = 4$

2 2

$F(x, y, z) = 2^3 = 8$

2 2 2

Logic Circuit

- Logic circuit is the hardware implementation of Boolean function using the logic gates.

- **Literals** – a variable or its complement in the function.

  e.g., $F(x,y,z) = x.y.z'$ — 3 literals x, y, z'

  $F(A,B,C) = A'.C + \underbrace{B'.C}_{\text{term}}$ — 4 literals A', C, B', C

- Each **literal** in the function is an input to a logic gate and each **term** is implemented with the logic gate.

- Minimizing the number of literals and terms results in a simpler circuit by reducing the number of logic gates.

Laws of Boolean Algebra

THEOREMS	AND	OR
Identity law	$A \,.\, 1 = A$	$A + 0 = A$
Annulment law	$A \,.\, 0 = 0$	$A + 1 = 1$
Idempotent law	$A \,.\, A = A$	$A + A = A$
Complement law	$A \,.\, A' = 0$	$A + A' = 1$
Commutative law	$A \,.\, B = B \,.\, A$	$A + B = B + A$
Associative law	$A.(B.C) = (A.B).C$	$A+(B+C) = (A+B)+C$
Distributive law	$A.(B+C) = A.B+A.C$	$A+(B.C) = (A+B).(A+C)$
Absorption law	$A.(A+B) = A$	$A+(A.B) = A$
DeMorgans law	$(A.B)' = A' + B'$	$(A+B)' = A' \,.\, B'$
Double Negation	$\overline{\overline{A}} = A$	

Duality & Complement of a Function

Duality Principle : Interchange binary operator and identity element
Imagine a Boolean function of 2 variables, **F(A,B) = A'.B + A.B'**

$$F_{dual} = (A' + B).(A + B')$$

AND : convert to OR , OR : convert to AND
1: convert to 0 , 0: convert to 1
No change to variables

Complement : derive using DeMorgans law

$$\mathbf{F'} \text{ or } \bar{\mathbf{F}} = (A + B')(A' + B)$$

AND : convert to OR , OR : convert to AND
1: convert to 0 , 0: convert to 1
Complement all variables

Simplify the expressions?

- $x + xy = x$

$x + xy = x.1 + xy$ ..... identity law, x.1 =x

$= x.(1 + y)$ ..... distributive law

$= x.\ 1$ ..... annulment law , y+1=1

$= x$ ..... identity law, x.1 =x

x	y	xy	x+xy
0	0	0	0
0	1	0	0
1	0	0	1
1	1	1	1

- $x + x'y = x+y$

$x + x'y = x + xy + x'y$ ..... absorption law, x+xy = x

$= x + y(x + x')$ ..... distributive law

$= x + y.1$ ..... complement law, x+x' = 1

$= x + y$ ..... identity law, y.1 = y

Summary

- Basics of logic design called Boolean Algebra.
- Laws and rules to define the operations.

AMRITA VISHWA VIDYAPEETHAM | AHEAD Online

Reference

- M Morris Mano - Computer System Architecture - PHI - Third Edition
- Gideon Langholz, Abraha& Joe L Mott - Digital Logic Design - World Scientific Publishing Co Ltd
- Thomas C Bartee - Digital Computer Fundamentals - Tata Mc Graw Hill - Sixth Edition

Objective

- Basics of Logic Gates

Logic gate

- Building blocks of digital electronic circuits.
- Boolean functions are implemented in digital circuits using these **logic gates**.
- Most logic gates have 2 inputs and 1 output.
- At any time, every terminal will be in one of the two binary conditions LOW (FALSE; 0;0V) or HIGH (TRUE;1;+5V).
- The function of each logic gate will be represented by Boolean expression

Logic gate

- Logic gates are classified into:

  Basic gates : OR, AND, NOT

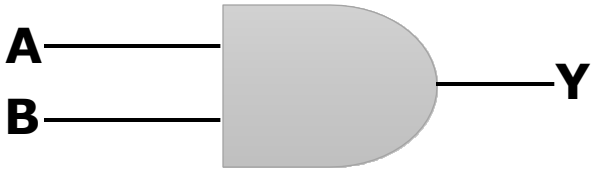
  Universal gate : NAND, NOR

  Special purpose gates: EX-OR, EX-NOR

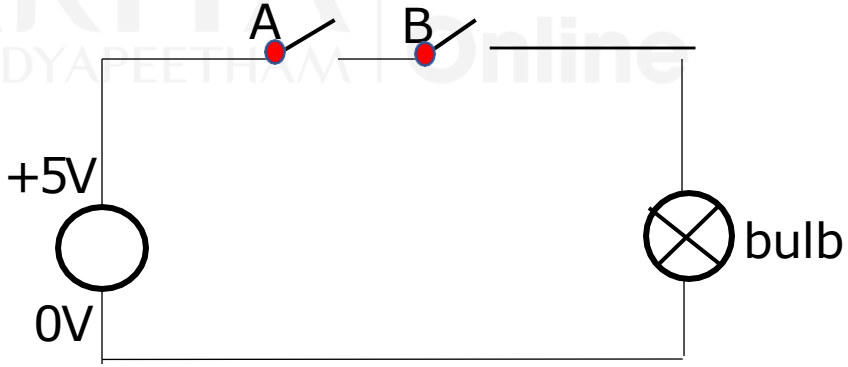
- Truth table explains all the outputs of a logic circuit for all possible inputs to that circuits.

AND gate

- The realization of logical AND operator .
- Logic expression, **Y =A.B** or **AB**
- The circuit will give high output (1) if both inputs are high otherwise, the output is low



Logic symbol



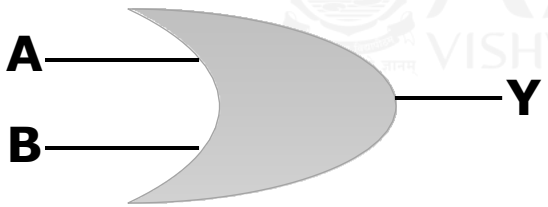
Switching circuit

inputs		output
A	**B**	**Y**
0	0	0
0	1	0
1	0	0
1	**1**	**1**

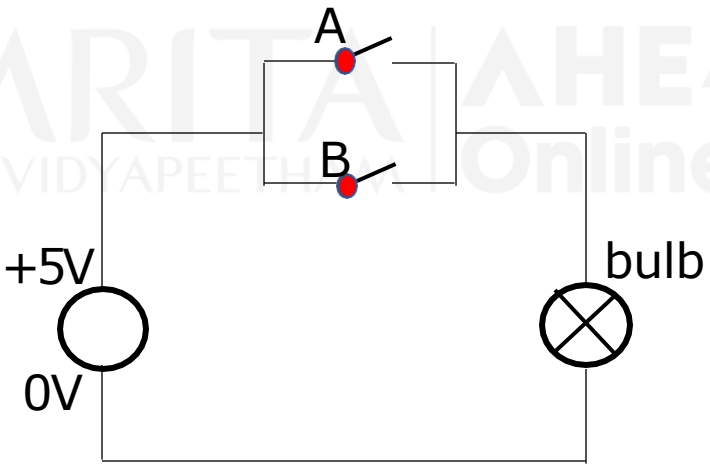
Truth table

OR Gate

- Logic expression, **Y=A+B**
- The circuit will give high output if any one input is high otherwise the output is low.



Logic symbol



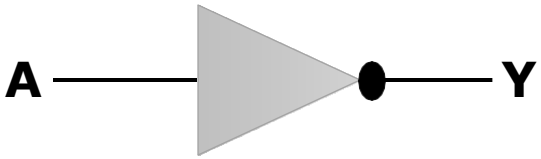
Switching circuit

inputs		output
A	**B**	**Y**
0	0	0
0	1	1
1	0	1
1	1	1

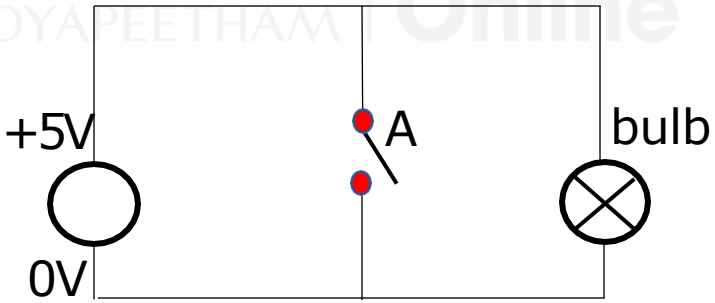
Truth table

NOT Gate

- Single input single output gate.
- Also called Inverter because it inverts the given binary input.
- Logic expression, **$Y=A' = \overline{A}$**



Logic symbol



Switching circuit

input	output
A	**Y**
0	1
1	0

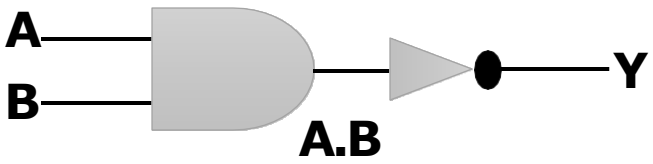
Truth table

NAND Gate

- NAND is a combination of AND gate and NOT gate
  NOT + **AND** = **NAND**

- The circuit will give high (1) output if any one input is low (0) otherwise, the output is low (0).

- Logical expression, $Y = \overline{A.B}$



or

inputs		output
A	**B**	**$Y=\overline{A.B}$**
0	0	1
0	1	1
1	0	1
1	1	0

Truth table

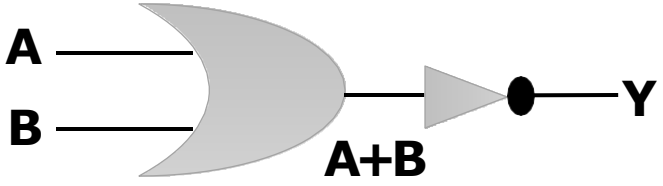
NOR Gate

- NOR is a combination of OR gate and NOT gate.

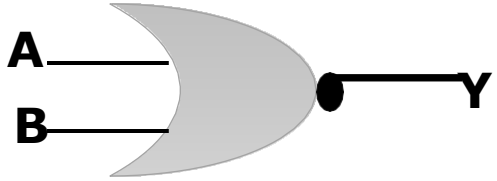
  NOT + **OR** = **NOR**

- The circuit will give high (1) output if both inputs is low (0) otherwise the output is high (1).

- Logical expression, $Y = \overline{A+B}$



or

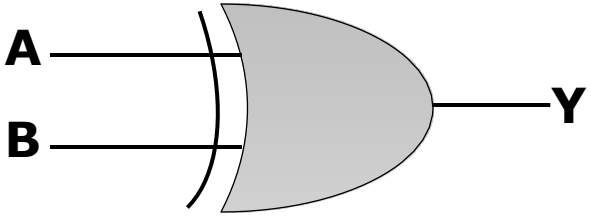


inputs		output
A	**B**	**$Y=\overline{A+B}$**
0	0	1
0	1	0
1	0	0
1	1	0

Truth table

XOR Gate

- Exclusive-OR gate
- The output is high if the circuit input has odd number of 1's otherwise the output is low i.e., with even number of 1's.
- Logical expression $\mathbf{Y = A \oplus B = A'B + AB'}$ (SOP)



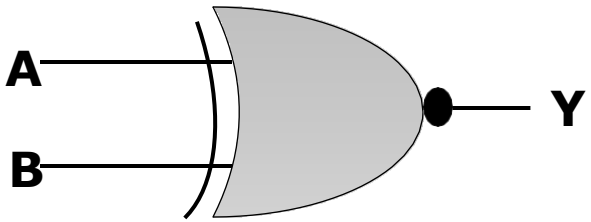
inputs		output
A	**B**	**Y**
0	0	0
0	1	1
1	0	1
1	1	0

Truth table

X-NOR Gate

- Exclusive NOR gate or equivalence.
- The output is high if the circuit input has even number of 1's otherwise the output is low.
- Logical expression $\mathbf{Y = A \odot B = A'B' + AB}$

Complement of XOR



inputs		output
A	**B**	**Y**
0	0	1
0	1	0
1	0	0
1	1	1

Truth table

Realizing Logic circuit from function

- Boolean Function, $\mathbf{F = x + y'z}$

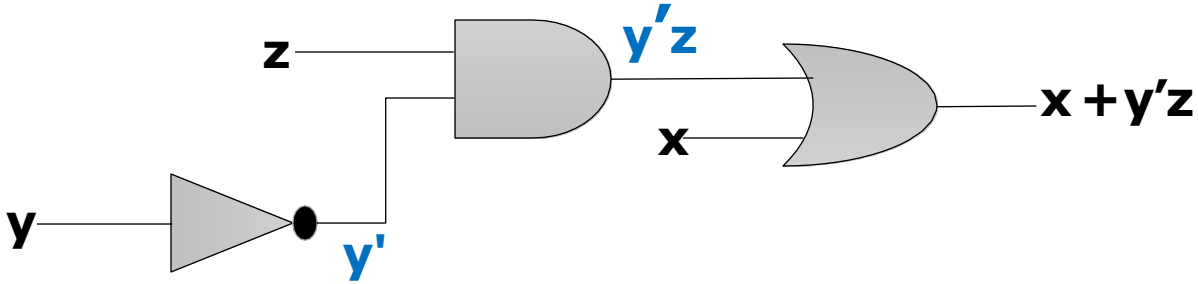
  input variable : x, y, z

- Truth table

  3 input variables = 2^3 combinations

- Logic circuit

  literals – 3 (x, y', z) and 2 terms



inputs			output
x	y	z	F=x+y'z
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

$x = 0, y = 1, z = 0$

$F = 0 + 1'.0$

$= 0 + 0.0$

$= 0 + 0$

$= 0$

$x = 1, y = 1, z = 0$

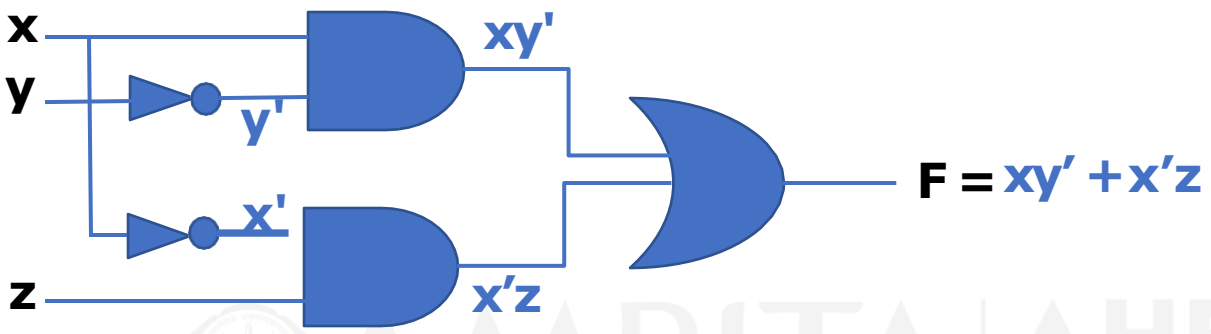
$F = 1 + 1'.0$

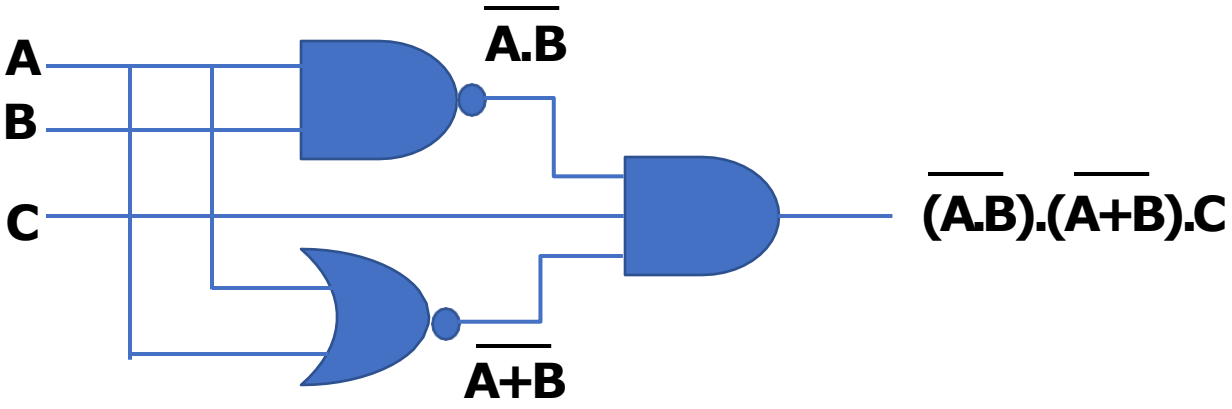
$= 1 + 0.0$

$= 1 + 0$

$= 1$

Realizing function from Logic circuit





Summary

Basics of Logic gates

Circuits

Combinational circuit

- The input values explicitly determine the output

Sequential circuit

- The output is a function of the input values and the existing state of the circuit

Circuit operations are described using

- Boolean expressions
- Logic diagrams
- Truth tables

Introduction

- Logic circuits for digital systems can be combinational or sequential.
- Combinational circuits consist of input variables, logic gates and output variables.
- Outputs are determined from the present input states not on previous states.

n inputs -2^n input combinations

For each input combination, only one possible output.

Combinational Circuits

- Adders
- Subtractors
- Comparators
- Decoders
- Encoders
- Multiplexers

Adder

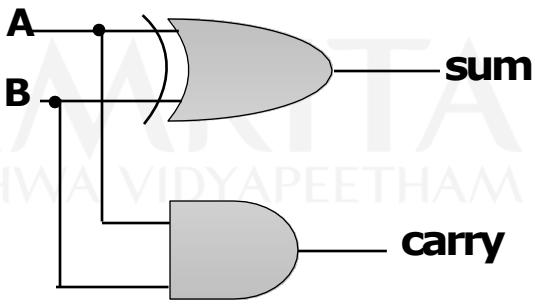
- Basic arithmetic operation in digital computer is binary addition.
- The combinational circuit which performs the addition of binary bits are called **adder**.
- **Half-adder** : Circuit performing the addition of 2 bits.
- **Full-adder** : Circuit performing the addition of 3 bits.
- Binary addition rules : **0+0 = 0 ; 0+1 = 1 ; 1+0 = 1 ; 1+1 = 0 with carry 1**

Half Adder

- 2 binary inputs augend and addend bits with 2 binary output called sum and carry.

inputs		outputs	
A	B	sum	carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Truth table



Logic diagram

Block diagram

Reference

- M Morris Mano - Computer System Architecture - PHI - Third Edition
- Gideon Langholz, Abraha& Joe L Mott - Digital Logic Design - World Scientific Publishing Co Ltd
- Thomas C Bartee - Digital Computer Fundamentals - Tata Mc Graw Hill - Sixth Edition